

Aryan Amarbhai Bijva

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Research Interests

Foundation model adaptation for dense regression in medical imaging, geometric deep learning for measurement tasks, graph neural networks for anatomical structure modeling, and robust AI for clinical deployment.

Education

Government Engineering College, Patan

Sept 2023 – June 2027 (Expected)

Bachelor of Engineering in Computer Science & Engineering

CGPA: 8.74 / 10.0

Relevant Coursework: Linear Algebra, Multivariable Calculus, Probability & Statistics, Design & Analysis of Algorithms, Deep Learning, Computer Vision, Medical Image Analysis.

Research Grants & Awards

CDAC GPU Grant

Nov 2025 – Present

MeitY, Government of India

National Level

Awarded high-performance compute resources (NVIDIA A100/H100 clusters) for medical foundation model training.

First student in institutional history to secure a national-level GPU grant for AI research.

Publications & Submissions

A. A. Bijva et al., “Template Overfitting in Foundation Model Adaptation for Dense Regression Tasks,” *submitted to Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026. [Under Review]*

Research Experience

Undergraduate Researcher

March 2025 – Present

Foundation Model Adaptation for Medical Regression

Identified **template overfitting** in foundation models (RETFound) adapted for dense regression, exhibiting 76% variance collapse in clinical predictions.

Developed a gradient-based loss function achieving 28% variance recovery (σ : $4.8\mu\text{m} \rightarrow 11.8\mu\text{m}$) for personalized diagnostics.

Architected a **Geometry-aware Graph Neural Network** with physical coordinate encoding and quantile regression for robust variance preservation.

Evaluated on the **FairFedMed** multi-site ophthalmic dataset (1,000+ scans) and cross-institutional validation using the GRAPE dataset.

Research Project

Jan 2025 – March 2025

Glaucoma Screening System Development

Developed automated glaucoma detection using fundus photography, achieving 92% accuracy on the REFUGE Challenge dataset.

Engineered an optic disc and cup segmentation pipeline for automated cup-to-disc ratio (CDR) computation.

Optimized model weights for inference on resource-constrained devices in rural healthcare settings.

Technical Skills

ML & Deep Learning: Foundation Models (ViT, RETFound), Graph Neural Networks (GNNs), Dense Regression, Geometric Deep Learning, Domain Adaptation, Quantile Regression.

Medical Imaging: Fundus Photography, OCT Analysis, DICOM/NIFTI Processing, Retinal Biomarker Extraction, Clinical Validation Metrics.

Scientific Computing: Python (PyTorch, PyTorch Geometric, NumPy, SciPy), MATLAB, L^AT_EX.

Infrastructure: Linux/HPC Cluster Management, NVIDIA A100/H100 Optimization, Git, Weights & Biases, MONAI.